Machine Learning Interview Questions

Career preparation notes Sources: standard material, listed under References

What this file is really about:

$$\text{what you are asked} \;\rightarrow\; \text{a precise answer} \;\rightarrow\; \text{the follow-up they will ask next}$$

How to use this file. Cover the answer, say yours aloud, then compare. Answer in this order: the idea, the formula, one example, and one limitation.

Step-up path

You need. The core machine learning sections.

1. Answer the fundamentals precisely.
2. Derive the standard results.
3. Discuss models and their limits.
4. Implement classics from scratch.
5. Handle the follow-up questions.

You can stop when you can answer standard technical questions under pressure.

Contents

Part A Fundamentals

A1 Learning and evaluation

What is the bias-variance trade-off? Expected squared error splits into bias squared, variance and noise. Simple models have high bias, flexible models high variance. Follow-up: bagging mainly reduces variance, and boosting mainly reduces bias.

Training error is low, test error high. What do you do? It is overfitting. Get more data, regularise, simplify the model, or use ensembling, and first check for leakage and data mismatch.

Accuracy is 99%. Is the model good? It depends on the base rate. With 1% positives, predicting all negative gets 99%. Report precision, recall, and PR-AUC.

ROC-AUC or PR-AUC? ROC-AUC is insensitive to class balance, so it can look good on rare events. PR-AUC reflects performance on the rare class.

What is data leakage? Information unavailable at prediction time reaches training, as with target encoding on all the data, or scaling before splitting. Fit preprocessing inside each fold.

How should you split time-series data? By time, using rolling origins. Random splits train on the future.

A2 Linear and logistic models

Derive the least-squares solution. Minimise $\|y - X\theta\|^2$. The gradient $-2X^\top(y - X\theta)$ set to zero gives $\theta = (X^\top X)^{-1}X^\top y$.

Why does L1 give sparse weights and L2 does not? The L1 ball has corners on the axes, so solutions land where weights are zero. Algebraically, the L1 subgradient gives soft thresholding.

Why not use squared loss for classification? With a sigmoid, it is non-convex and gives vanishing gradients on confident errors. Log loss is convex and punishes confident mistakes.

What does a logistic regression coefficient mean? A one-unit increase adds w to the log-odds, and multiplies the odds by e^w.

How do correlated features affect linear regression? Predictions stay fine, while coefficients become unstable.

A3 Trees and ensembles

How does a tree choose a split? The split that most reduces impurity, using Gini or entropy for classes, and variance for regression.

Random forest or boosted trees? Forests are robust with little tuning, and reduce variance. Boosting usually scores higher, but needs tuning and can overfit noise.

How does gradient boosting work? Each new tree fits the negative gradient of the loss at the current predictions.

How does XGBoost differ? It uses second derivatives, and regularises tree size and leaf weights.

Are tree feature importances trustworthy? Impurity importance favours high-cardinality features. Prefer permutation importance on held-out data.

A4 Other classics

What is the kernel trick? Replacing dot products with a kernel gives the effect of high-dimensional features, without computing them.

What are support vectors? Points on or inside the margin. Only these determine the boundary.

K-means assumptions and failures? Round, similar-sized clusters. It fails on rings, moons and strongly unequal groups.

PCA versus t-SNE? PCA is linear and preserves variance, with interpretable axes. t-SNE preserves neighbours, and its global distances are not meaningful.

Why standardise features? For distance- and gradient-based methods. Trees do not need it.

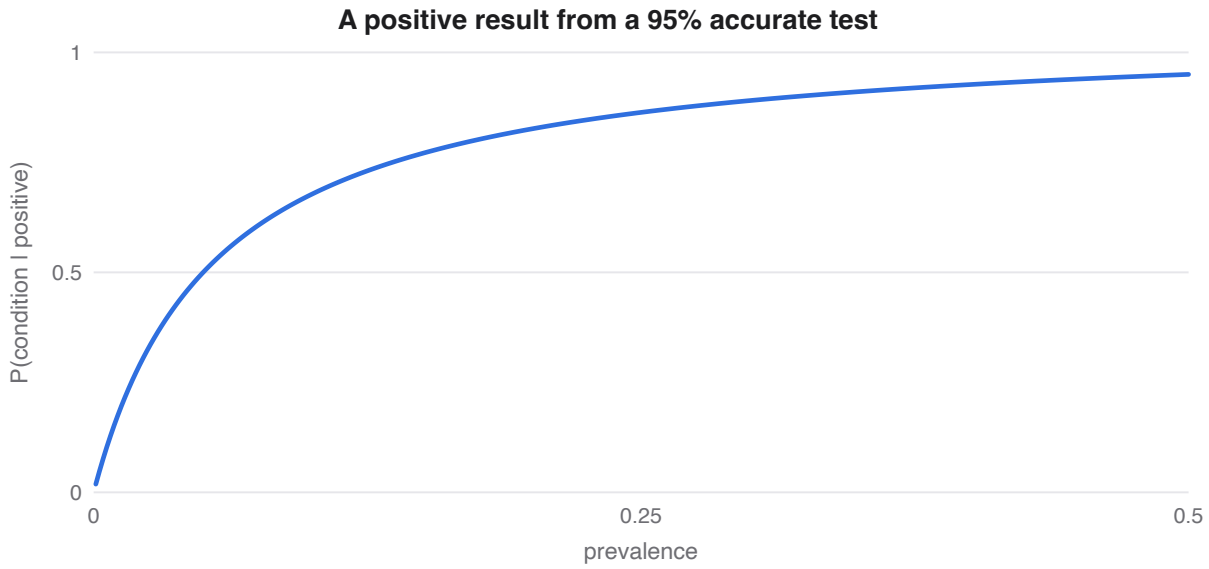
A5 Probability and statistics

State Bayes' rule, and give a use. $P(A \mid B) = P(B \mid A)P(A)/P(B)$. A rare disease with a good test still gives mostly false positives.

MLE versus MAP? MLE maximises the likelihood. MAP adds a prior, and a Gaussian prior is ridge.

What does a p-value mean? The probability, under the null, of data at least this extreme. It is not the probability the null is true.

What is the central limit theorem? Sums of independent variables with finite variance tend to a normal distribution, so sample means are approximately normal.



A6 Neural networks and deep learning

Why are non-linearities needed? Otherwise a stack of layers is one linear map.

Explain backpropagation. Reverse-mode automatic differentiation: one backward sweep applying the chain rule gives all gradients.

What causes vanishing gradients, and what are the fixes? Multiplying many small derivatives. Fixes are ReLU, residual connections, normalisation and good initialisation.

What does batch norm do? It normalises activations using batch statistics, which allows higher learning rates. It behaves differently at training and inference.

Why does dropout work? It prevents co-adaptation, acting like averaging many thinned networks.

Adam or SGD? Adam converges quickly with little tuning. SGD with momentum can generalise slightly better on some vision tasks.

How does attention work? Each token forms a query against every key, and a softmax over their scores mixes the values.

Why do transformers need positional information? Attention treats tokens as a set.

What is the difference between a CNN and a ViT? A CNN has locality and translation equivariance built in, while a ViT learns relationships and needs more data.

How does diffusion work? Add noise step by step, train a network to predict that noise, and generate by removing it from pure noise.

What are hallucinations, and how do you reduce them? Confident invented output. Use retrieval, require citations, and evaluate groundedness.

Part B In Code

B1 Implement K-means

```python
import numpy as np

def kmeans(X, k, iters=100, seed=0):
    rng = np.random.default_rng(seed)
    C = X[rng.choice(len(X), k, replace=False)]
    for _ in range(iters):
        labels = ((X[:, None] - C[None]) ** 2).sum(-1).argmin(1)
        new = np.array([X[labels == j].mean(0) if (labels == j).any() else C[j] for j in ra
        if np.allclose(new, C):
            break
        C = new
    return C, labels
```

B2 Implement logistic regression

```python
def fit_logistic(X, y, lr=0.1, steps=1000):
    X = np.c_[np.ones(len(X)), X]
    w = np.zeros(X.shape[1])
    for _ in range(steps):
        p = 1 / (1 + np.exp(-X @ w))
        w -= lr * X.T @ (p - y) / len(y)
    return w
```

B3 Implement a decision stump

```
def gini(y):
    p = np.bincount(y) / len(y)
    return 1 - (p ** 2).sum()

def best_split(x, y):
    best = (None, np.inf)
    for t in np.unique(x)[:-1]:
        left, right = y[x <= t], y[x > t]
        score = (len(left) * gini(left) + len(right) * gini(right)) / len(y)
        if score < best[1]:
            best = (t, score)
    return best
```

B4 Precision, recall and AUC from scratch

```
def precision_recall(y, pred):
    tp = ((pred == 1) & (y == 1)).sum()
    return tp / max(pred.sum(), 1), tp / max(y.sum(), 1)

def roc_auc(y, score):
    order = np.argsort(score)
    ranks = np.empty(len(y)); ranks[order] = np.arange(1, len(y) + 1)
    n1, n0 = y.sum(), len(y) - y.sum()
    return (ranks[y == 1].sum() - n1 * (n1 + 1) / 2) / (n1 * n0)
```

Part C Practice

P1. Precision 0.5, recall 1. Find the F1 score.

$2 \times 0.5 \times 1 / 1.5 = 0.667$.

P2. A model with 100 features has perfect training accuracy on 50 rows. What would you suspect?

Overfitting, since there are more parameters than examples. Check validation performance, and regularise.

P3. What is the expected number of unique items in a bootstrap sample of size n?

About $0.632n$.

P4. A 3-class problem has a starting loss of 1.10. Is it healthy?

Yes. $\ln 3 = 1.10$.